



RAISE-Bot

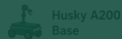
— Hands-on ROS 2, AI Agents, and Greenhouse Robot Inspection



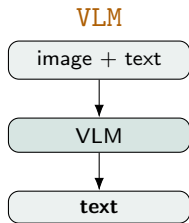
Comparative VLA Models

Student edition — which VLA for the greenhouse robot?

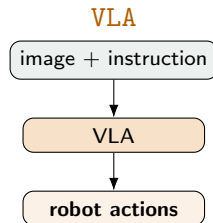
Professor Anis Koubaa • 14–16 July 2026



Day 1 the robot *thinks*; Day 2 it *acts*



perception & description only



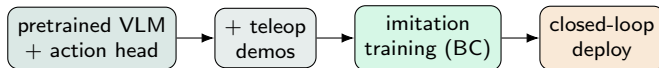
predicts the motor commands itself

A **VLA** maps pixels + language **directly to actions** — no controller in between. Day 2: pick one and make it move the UR5e.

How do you *build* a VLA? — the 5-step recipe

The recipe (any VLA)

- 1 Start from a **pretrained VLM** — it already sees & reads.
- 2 Bolt on an **action head** — action *tokens*, or a flow/diffusion expert.
- 3 **Collect demonstrations** (teleop): image + state + action, at ~ 30 Hz.
- 4 **Train by imitation** (behavioral cloning): predict the demonstrated action.
- 5 **Deploy closed-loop**: observe $\rightarrow$ action chunk $\rightarrow$ execute $\rightarrow$ repeat.



The hard part is **steps 3–4**: **good demonstrations** matter more than a bigger model.

Our greenhouse example

- 1 **SmolVLM2** backbone (vision + language).
- 2 flow-matching expert $\rightarrow$ **UR5e** joint actions.
- 3 30–50 teleop *tomato-pick* episodes (LeRobot v2.0).
- 4 fine-tune **smolvla_base** on them ($\sim 20k$ steps).
- 5 “*pick the ripe red tomato*” $\Rightarrow$ the arm moves.

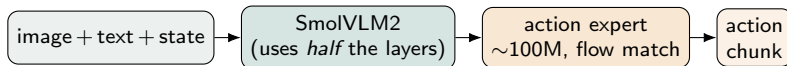
The open-source contenders (mid-2026)

- **OpenVLA** (Stanford, 2024) — the 7B reference; strong, but heavy. **OpenVLA-OFT** (2025) makes it fast.
- π_0 / $\pi_{0.5}$ (Physical Intelligence, 2025) — current SOTA generalist; flow-matching, smooth high-rate actions.
- **SmolVLA** (Hugging Face / LeRobot, 2025) — 450M, built for consumer hardware and **our exact data format**.
- **GROOT N1.5** / **N1.7** (NVIDIA, 2025) — 3B, humanoid-focused, Isaac.
- **ACT** / **Diffusion Policy** — not VLAs (no language) but tiny, reliable *baselines* that train in minutes.

*All open-weight. The question is which one fits a **single GPU + a 90-minute class**.*

SmolVLA — the efficient, LeRobot-native VLA (2025)

- ① **Started** Hugging Face / LeRobot, Jun 2025. Goal: a VLA that runs on **consumer hardware** (even a MacBook).
- ② **Scope** Low-cost arms; trained on **~480 community LeRobot datasets** — **our exact recording format**.
- ③ **Objective** *Democratize VLAs* — big-model behaviour at 450M params, fine-tunable on one GPU.
- ④ **How it works** — a trimmed VLM + a small flow-matching expert:



Speed tricks: **skip half the VLM layers**, interleave self/cross-attention, and **async inference** (act while thinking). 82–90% LIBERO at $16\times$ fewer params.

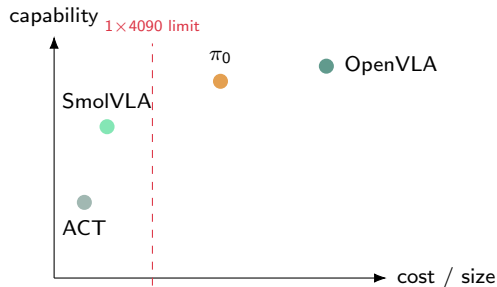
Side by side — the facts

Model	Params	Released	License	Pretrained on
SmolVLA	450 M	Jun 2025	Apache-2.0	~480 LeRobot community sets
π_0 / $\pi_{0.5}$	~3 B	Feb 2025	Apache-2.0	10k+ hours robot data
OpenVLA	7 B	Jun 2024	MIT	970k Open-X episodes (22 robots)
OpenVLA-OFT	7 B	Feb 2025	MIT	OpenVLA + fast OFT recipe
GR00T N1.5/1.7	3 B	2025	NVIDIA	humanoid data + sim
ACT / Diff. Policy	~80 M	2023–24	open	per-task (no language)

Highlighted row = our choice. *Bigger is not better when you have one GPU and one class period.*

Performance vs. cost — the trade we care about

- **SmolVLA**: **82–90%** on LIBERO — within $\sim 5\%$ of OpenVLA at **16 $\times$ fewer parameters**. Async inference: 30% faster, 2 $\times$ throughput.
- **OpenVLA**: strongest raw success, but base inference is slow. **OFT** fixes it: **109.7 Hz**, **0.073 s/chunk**, LIBERO 76.5 $\rightarrow$ 97.1%.
- π_0 : smooth ~ 50 Hz action streams; SOTA generalization.



Left of the red line = fine-tunes on our hardware.

The axis the others ignore: fits a single RTX 4090?

Model	Fine-tune on 1×4090 (24 GB)?	LeRobot-native?
SmolVLA	✓ easily (~12 GB, ~50 demos)	✓ built by LeRobot
π_0 / $\pi_{0.5}$	✓ LoRA only, tight	✓ (openpi loaders)
OpenVLA	✗ full FT needs an A100	partial
GR00T N1.5	heavy	✗ (Isaac, not LeRobot)
ACT / Diff. Policy	✓ trains in minutes	✓

We already record in **LeRobot v2.0 format**. SmolVLA consumes it **directly** — no embodiment-gap glue, no conversion.

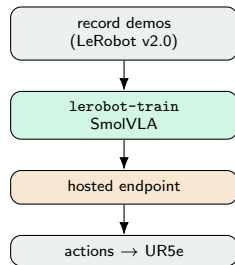
Our choice for RAISE: SmolVLA

Why it wins here:

- 1 **LeRobot-native** — consumes our recordings as-is.
- 2 **Fine-tunes inside a 90-min lab** on one 4090 (30–50 demos).
- 3 **Impressive enough** — 82–90% LIBERO, ~5% behind a 7B model.
- 4 **Real-time** — async inference beats our 500 ms/chunk budget.

Roles for the others:

- π_0 — advanced / "frontier" stretch goal.
- **OpenVLA(-OFT)** — demo & discussion of the 7B reference.
- **ACT** — **safety net** if a team's VLA won't converge in time.



Lab 2.1 feeds Lab 2.2.

Key Takeaways

- **VLA = image + instruction $\rightarrow$ actions** (no hand-written controller).
- **Bigger isn't better** under one GPU + one class hour — match the model to the constraint.
- **SmolVLA** is our pick: LeRobot-native, fine-tunes on a 4090, real-time.
- π_0 / **OpenVLA-OFT** for the frontier story; **ACT** as the reliable fallback.

Questions?

Next: *Lab 2.1 — teleoperate the arm & record a LeRobot dataset.*